



External suction-blowing method for controlling vortex-induced vibration of a bridge

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ARTICLE INFO

Keywords:

External suction-blowing method
Three-dimensional spanwise-varying control
Vortex-induced vibration
Bridge
Wind tunnel tests

ABSTRACT

To suppress the vortex-induced vibration (VIV) of long-span bridges, an active flow control method, i.e. the external suction-blowing method (ESBM), is proposed in this study based on the efficiency of the three-dimensional spanwise-varying flow control method and the suction/blowing flow control method. The method can be achieved by cyclically arranging suction and blowing on the undersurface of the bridge. The effectiveness of the method for the vertical and torsional VIV of the bridge sectional model is verified through wind tunnel tests. The test results imply that the best perturbation position of the ESBM is the leading edge of the undersurface of the bridge. For dimensionless spanwise distances of 2–4 with single-hole flow coefficients of 2.27×10^{-3} – 3.03×10^{-3} and 1.45×10^{-3} – 1.94×10^{-3} during vertical and torsional VIV, respectively, the VIV can be completely suppressed. The results of the wake analysis show that this method can effectively suppress the scale of the spanwise vortex of the bridge, causing the VIV to disappear.

1. Introduction

Vortex-induced vibration (VIV) in long-span bridges occurs in the low wind speed range. In this range, the wind flowing through the main girder will cause periodic vortex shedding in the wake, which will result in aerodynamic changes (Komatsu and Kobayashi, 1980). When the frequency of the vortex shedding is equal to the natural frequency of the structure, VIV occurs. This type of vibration is a common form of wind-induced vibration in long-span bridges, which will affect the safety of driving on these structures. While VIV is a self-limited vibration, it can cause long-term fatigue damage (Jim and Kim, 2017; Zhang et al., 2019).

To date, research on suppressing the VIV of long-span bridges has mainly focused on flow control methods. These flow control methods can be divided into active and passive methods based on whether external energy is required. Passive control methods are often considered in research owing to their simple structure and low cost. Larose et al. (2003) studied the effect of a deflector on the VIV control of Stonecutters Bridge and found that at low Reynolds numbers, the presence of the deflector would increase the VIV amplitude of the main beam, while at high Reynolds numbers, the amplitude could be reduced. Schewe and Larsen (1998) noted that the deflector had a significant

affect on weakening the VIV under zero wind attack angle. However, under a large wind attack angle ($+3^\circ$ or -3°), the VIV was enhanced at low Reynolds numbers, indicating that the deflector is only effective within a certain range. Shigeru and Fumoto (2008) studied the effect of central slotting methods on the three aerostatic coefficients of a bridge girder using computational fluid dynamics (CFD) simulations and observed the flow field around the main girder when the central slot was combined with a wind barrier. When the wind attack angle was greater than 12° , the wind barrier had a significant influence on the aerodynamic torque coefficient. Kwok et al. (2012) demonstrated in wind tunnel tests that the gap width of the twin box bridge and wind attack angle have a significant influence on the pressure distribution on the surface of the main girder. The entire downstream deck is located in the wake area of the upstream deck under a large gap width, and the drag coefficient increases significantly; the lift coefficient and torque coefficient of the main girder under different wind attack angles are clearly different.

Active control methods have developed rapidly in recent years owing to their flexibility and control efficiency. The actively controlled flap is the most advanced research direction for active flow control. Kobayashi and Nagaoka (1992) first proposed this method and attempted to suppress the flutter of bridges used actively controlled flaps. Using these

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<https://doi.org/10.1016/j.jweia.2021.104661>

Received 20 November 2020; Received in revised form 21 April 2021; Accepted 11 May 2021

Available online 3 June 2021

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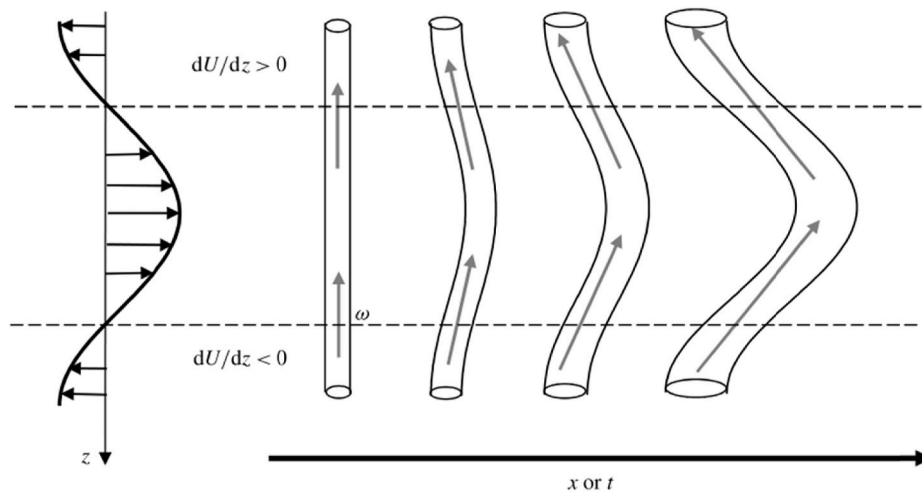


Fig. 1. Evolution of the secondary instability of a spanwise vortex (Hwang et al., 2013).

flaps had a positive effect and could double the critical wind speed of the flutter. Hansen et al. (2000) combined the real-time position and movement of the main girder and adjusted the flaps through a control algorithm to allow the wind force on the aerodynamic flaps to offset the vibration of the main girder. Boberg et al. (2015) proposed a control law for aerodynamic wings through wind tunnel tests and achieved very good results. The oscillations were suppressed within 0.5 s–5 s, which means that the stabilisation speed was three times faster under the proposed control law.

Based on the perturbation position, flow control methods can be divided into two-dimensional (2D) control methods and three-dimensional (3D) control methods. Most traditional flow control methods are 2D control methods. Kim and Choi (2005) compared the effects of 2D and 3D suction/blowing control systems on cylinders. The results showed that the 3D control method was more efficient than the traditional 2D control method. Because of this improved efficiency, many scholars have applied this method to the field of wind-induced vibration flow control of long-span bridges by analysing the resulting flow fields. Through numerical simulations and wind tunnel tests, Hyungmin et al. (2006) determined that placing small vertical plates with periodic intervals on the trailing edge of the bridge undersurface could effectively reduce aerodynamic drag. Deshpande and Sharma (2012) achieved a reduction in aerodynamic drag through periodic slotting at the undersurface of the sectional bluff. El-Gammal et al. (2007) transformed the leading and trailing edges of a bridge sectional model into a wave in the spanwise direction and completely suppressed the VIV in wind tunnel tests. Zhang et al. (2018) studied the effect of vortex generators on the wind-induced vibration of bridges through wind tunnel tests and found that periodically arranging vortex generators could completely suppress VIV.

In 3D spanwise-varying active flow control, many scholars have focused on the effects of suction and blowing on the flow field around a bluff body. Menicovich et al. (2014) studied the influence of jets on the flow fields of tall buildings in wind tunnel tests; the results showed that the active aerodynamic measures of jets affected the lift, drag coefficient, and frequency of vortex shedding in the wake area. The use of high-momentum jets resulted in a higher frequency of vortex shedding and a smaller response. Akansu and Erkan (2010) used Particle Image Velocimetry (PIV) to study the influence of the jet on the flow field of a rectangular body in wind tunnel tests. It was found that the jet would change the frequency of vortex shedding and the length of the recirculation zone as well as the average and fluctuating pressure at the trailing edge. Based on the influence of the jet on the flow field around a bluff body structure and the efficiency of 3D spanwise-varying control, flow control methods using suction or blowing have been proposed. Chen

et al. (2013) studied the control effect on VIV of cylinders obtained with a self-issuing jet, and the test results showed that this method could effectively reduce the amplitudes of the VIV and the fluctuating lift coefficient. Naghib-Lahouti et al. (2012) studied the effect of a 3D spanwise-varying flow control method on the trailing edge of a bridge through wind tunnel tests and determined that the spanwise-varying jet could effectively suppress the spanwise vortex of the bridge sectional model and reduce the aerodynamic drag. Zhang et al. (2016) verified that spanwise suction could effectively reduce the aerodynamic coefficient of a bridge sectional model through numerical simulations.

2D flow control methods based on cross-sectional analysis require a continuous arrangement of control devices along the spanwise direction, which is not economical. Moreover, the aerodynamic shape of the main girder can only be improved within a limited wind speed range, and a large number of tests and simulations are required to verify the effectiveness before use. The control mechanisms of the 3D and 2D control methods are different. Grant et al. (1958) determined that pairs of downstream vortices appeared in the wake of a cylinder, which is the secondary instability of the spanwise vortex. The principle of 3D spanwise-varying control methods is to use the secondary instability of the spanwise vortex and excite the downstream vortex by arranging spanwise variations at periodic intervals, which limits the development and shedding of the spanning vortex. Hwang et al. (2013) described the evolution of the secondary instability of the spanwise vortex, as shown in Fig. 1. The spanwise variation changes the downstream velocity distribution of the wake such that the time-averaged distribution of the velocity changes periodically, which causes the spanwise vortex to evolve into a pair of downstream vortices. There are three types of secondary instabilities in a bluff body wake: Mode-A, Mode-B, and Mode-C (Mode-S), where Mode-A changes the rotation direction of the downstream vortex twice during a spanwise vortex shedding cycle, Mode-B does not change the direction of rotation in a cycle, and Mode-C changes the direction of rotation once (Williamson, 1996; Lewke and Williamson, 1998; Choi and Yang, 2014).

The advantages of 3D spanwise-varying flow control methods and suction/blowing methods have become of increasing research interest, and their effectiveness has gradually been recognised. If suction and blowing occur cyclically in the spanwise direction of the undersurface of the main girder as a result of the control device, it may become an effective method for controlling the VIV of the bridge main girder. Based on this concept, a flow control method called the external suction-blowing method (ESBM) is proposed. In this study, wind tunnel tests are used to investigate the control effect and mechanism of this method on the VIV of a bridge sectional model.

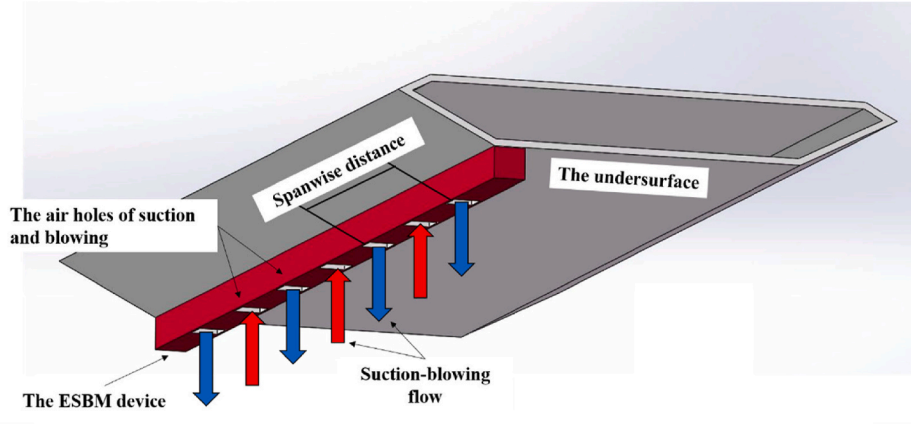


Fig. 2. ESBM device and parameter definitions.

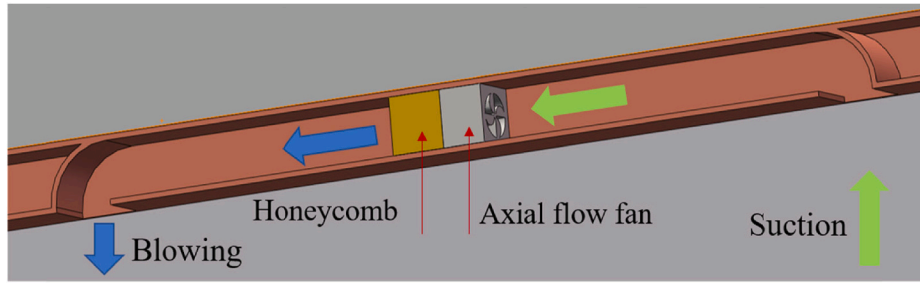


Fig. 3. Working principle of the ESBM device.

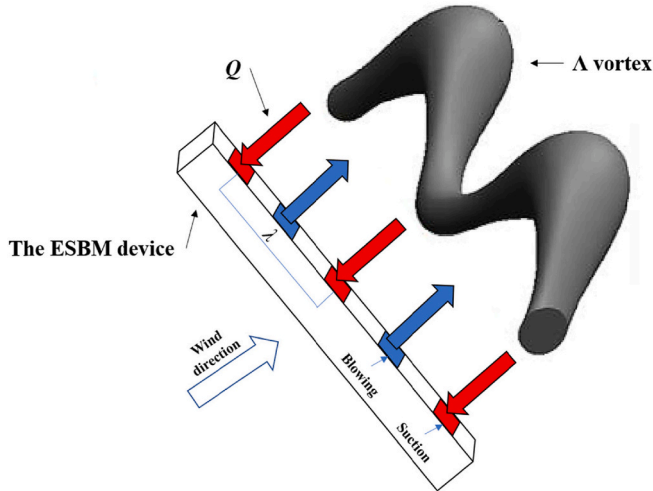


Fig. 4. Control principle of the ESBM.

2. External suction-blowing method

The ESBM is a 3D spanwise-varying flow control method. The perturbations are supplied by periodically arranged suction and blowing perturbations along the spanwise direction. The size of the external devices for providing the perturbations should not be large to avoid affecting the flow field characteristics of the bridge and increasing the wind-induced vibration. The device is a long square tube with independent closed spaces located periodically in the spanwise direction. The device is installed on the bottom of the bridge, and the air holes on the bottom of the tube are used as the suction and blowing outlets. The air source is provided by an axial flow fan installed in the centre of the

independent space inside the tube. The positive and negative pressures before and after the fan are used to achieve the effect of suction at one end of the tube and blowing at the other. The device and parameter definitions are shown in Fig. 2. The working principle of the device is shown in Fig. 3.

The control principle of the ESBM for the VIV of long-span bridges, which is shown in Fig. 4, is the same as the control principle of 3D spanwise-varying methods. By applying the perturbation of suction and blowing, the secondary instability of the large-scale spanwise vortex is excited, and the downstream vortex suppresses the development of the spanwise vortex, reducing the scale of the spanwise vortex, and thus suppressing the VIV of the main girder.

Considering the installation and effect of the device on the flow field, a square cross-sectional shape of the tube is selected. A single-hole flow coefficient equation can be defined as follows:

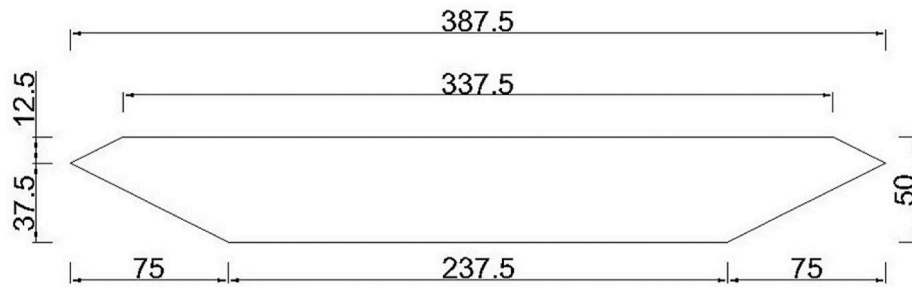
$$C_Q = \frac{Q}{UBH} \quad (1)$$

where Q is the single-hole flow, U is the wind speed corresponding to the maximum VIV amplitude of the girder under no control state, B is the width of the girder, and H is the height of the girder. Because the single-hole flow coefficient of the bridge sectional model is equal to that of the real bridge, combined with the reduced wind speed equation, the equation for the real bridge flow can be given as follows:

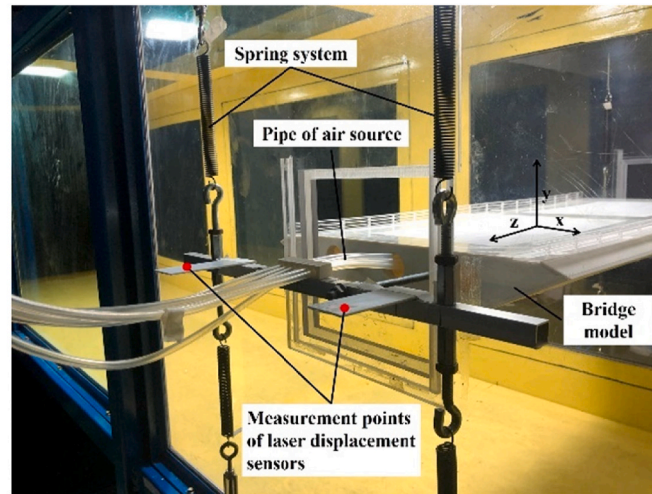
$$Q = \frac{f}{f_0 s^3} Q_0 \quad (2)$$

where s is the geometric scale ratio, f is the natural frequency of the real bridge, H_0 is the height of the model, B_0 is the width of the model, and Q_0 is the effective flow of the model.

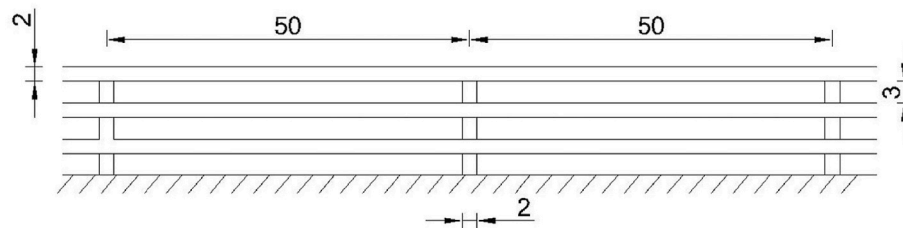
The main parameters in this method are the single-hole flow coefficient and spanwise distance, which are normalised by the height of the main girder. As can be seen from Fig. 4, λ is the spanwise distance



(a) Size of the bridge sectional model (units: mm)



(b) Suspension system



(c) Size of the railing (units: mm)

Fig. 5. (a) Size of the test model, (b) photograph of the suspension system, and (c) size of the railing.

between adjacent suction (or blowing) holes of the bridge section model, and the dimensionless spanwise distance is defined as $\Lambda = \lambda / H$.

When the ESBM is applied to a real bridge, by designing a circuit in the tubes and driving the fan with electrical energy, the suction and blowing flow can be produced, aim at suppressing bridge vibration.

3. Experimental setup

These experiments were conducted in a wind tunnel at the Northeast Forestry University in China. A closed reflow wind tunnel with a rectangular cross section of $1 \text{ m} \times 0.85 \text{ m}$ was used; the test section is 6 m long. The wind speed in the wind tunnel is continuously adjustable in the range of 2–70.5 m/s, the turbulence intensity is less than 0.5%, and the non-uniformity is less than 1%. The bridge sectional model responses included the vertical displacement, torsional displacement, and downstream wind speed in the wake region. The displacement signals were measured using three laser displacement sensors placed at both ends of the model and aimed at targets located on the rigid links. The sampling

frequency of the laser displacement sensors was 1000 Hz. The downstream wind speed signal was measured with a hot-wire anemometer located behind the model. The sampling frequency of the hot-wire anemometer was 1000 Hz. For each wind speed, 1 min of data was collected after the wind speed and model vibration had stabilised.

The prototype for the model is the Great Belt Bridge East Bridge in Denmark. The model, as shown in Fig. 5 (a), is constructed of ABS plastic with a thickness of 3 mm. The 1:80 scale model has a spanwise length $L = 0.8 \text{ m}$, width $B = 0.3875 \text{ m}$, and height $H = 0.05 \text{ m}$. The end plate of the model is equipped with bolts to connect the rigid links. The slot on the undersurface is used to install the ESBM device, and an enlarged hole is located on the top plate to make installation of the device more convenient. The end plate is provided with holes to allow the gas source pipeline to enter the model and connect to the device. The connection method between the model and the free suspension system is shown in Fig. 5 (b), where the blockage ratio is approximately 5%. The mass and mass moments of the model are $m = 15.281 \text{ kg/m}$ and $I = 0.449 \text{ kg m}^2/\text{m}$, respectively. The non-dimensional values are $m/\rho B^2 = 21.83$ and $I/$

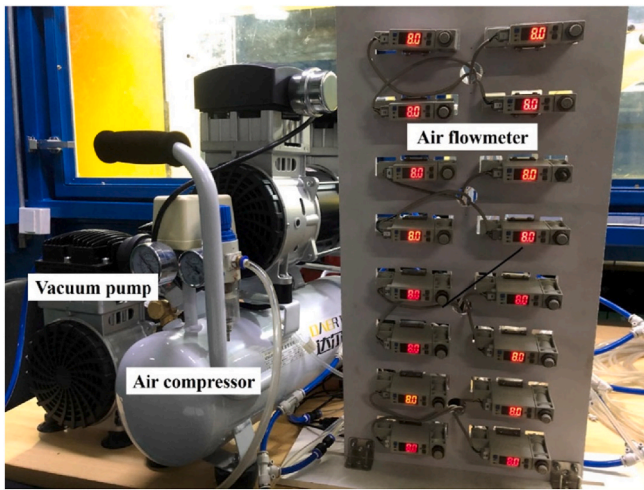


Fig. 6. Air source and flow monitoring system.

$\rho B^4 = 15.4$, where $\rho = 1.226 \text{ kg/m}^3$ is the air density, and B is the model width. The coil springs are mounted symmetrically on the rigid links to provide vertical and torsional stiffness. The vertical and torsional frequencies of the test model are $f_v = 4.52 \text{ Hz}$ and $f_t = 6.90 \text{ Hz}$, respectively, and the corresponding damping ratios are $\xi_v = 0.37\%$ and $\xi_t = 0.31\%$. Railings, which are the main structure on long-span bridges, will increase the VIV response of the bridge. The shape and size of the railing model used in these tests are shown in Fig. 5 (c).

The device designed for these wind tunnel tests is based on the Great Belt Bridge East Bridge in Denmark. Based on experience, the effective flow of the model is 8 L/min , which is similar to the effective suction flow used by Zhang et al. (2016). The geometric scale ratio and model size are as described above, and the wind tunnel tests simulate the vertical VIV frequency of real bridge is $f = 0.209 \text{ Hz}$ (Larsen et al. (2000)). Therefore, according to the reduced wind speed formula, the wind speed during maximum vertical VIV amplitude of real bridge is $U = 8.42 \text{ m/s}$. The calculation shows that the flow of the tubes is approximately $11364 \text{ m}^3/\text{h}$. When the length of the square tube section is 0.5 m , the corresponding flow velocity is 12.63 m/s which meets the flow velocity range of industrial ducts is $6\text{--}14 \text{ m/s}$ in Chinese Standard (Code for acceptance of construction quality of ventilation and air conditioning engineering (GB 50243–2016)). The length of the device model was calculated according to the scale ratio as 6.25 mm . Because the size of the device is so small, no variable-frequency axial flow fan matching its size was available. Therefore, a vacuum pump was used to provide the suction air source, and an air compressor was used to provide the blowing air source. This method can replace the fan equivalently. The flow rate was adjusted by a flow meter to achieve a stable flow rate. The entire system connects the air source system, flow monitoring system, and device through soft tubing. The air source and flow monitoring systems are shown in Fig. 6.

The main control variables in these experiments were the perturbation mode, dimensionless spanwise distance, and single-hole flow coefficient. To ensure that the presence of the external device would not reduce the original wind-induced vibration stability of the bridge, it was necessary to verify the control effect of the placement of the device on the response of the VIV. Combining the placement of the device and the application of suction-blowing action, the effect of different perturbation modes for controlling the VIV of the bridge sectional model were discussed. The perturbation modes are shown in Fig. 7.

As can be seen from Fig. 7 (i) and (g), the device is installed on the leading and trailing edges of the undersurface of the model at the same time. In the same section, the air holes on leading and trailing edges are in the same state, both suction or blowing, which is defined as FB-mode, and the air holes on leading and trailing edges are in the opposite state,

one suction and the other blowing, which is defined as FBX-mode.

As a 3D spanwise-varying control method, the dimensionless spanwise distance of the ESBM is the key to the excitation of the optimal mode of the secondary wake instability. In these experiments, the dimensionless spanwise distance was set as $\Lambda = 2\text{--}5$.

The difference with the application of flow control is very obvious, and thus it is necessary to explore the effect of different single-hole flow coefficients (C_Q) for controlling the VIV of the bridge sectional model. In these experiments, the single-hole flow coefficients were set at $0.76 \times 10^{-3}\text{--}3.03 \times 10^{-3}$ for the vertical VIV and $0.48 \times 10^{-3}\text{--}1.94 \times 10^{-3}$ for the torsional VIV (at 2 L/min intervals, the flow rate was $2\text{--}8 \text{ L/min}$).

The control effect of the ESBM under different wind attack angles (α) was also discussed. The wind attack angles were set at $\pm 3^\circ$.

The model vibration response is represented by plots of the root mean square (RMS) of the vertical and torsional displacement and the non-dimensional wind speeds, which are $U/f_v B$ and $U/f_t B$ (U is the wind speed, $f_v = 4.52 \text{ Hz}$ and $f_t = 6.90 \text{ Hz}$ are the vertical and torsional natural frequencies, respectively, and $B = 0.3875 \text{ m}$ is the model width). The vertical RMS is normalised by the model height, H . The wake evolution of the model is obtained from the downstream wind speed measured using a hot-wire anemometer. The wind speed parameters are the mean downstream wind speed and the fluctuating downstream wind speed, which are non-dimensionalised based on the wind speed of the maximum amplitude.

4. Results and discussion

(1) Effects of the perturbation mode on control of the VIV

Under $\alpha = 0$, $\Lambda = 3$, $C_Q = 3.03 \times 10^{-3}$ for vertical VIV and $C_Q = 1.94 \times 10^{-3}$ for torsional VIV, the control effects of the ESBM devices for reducing the vortex-induced vertical and torsional responses with different perturbation modes are shown in Fig. 8.

It can be seen from the results that different placements of the device have a greater influence on the VIV response when the flow is uncontrolled. In the plot of the vertical response, compared with the no control state, it can be seen from the change in the RMS of the vertical displacement that when the ESBM device is installed on the trailing edge of the undersurface, the vibration response increases, which is an adverse effect. This demonstrates that the trailing edge of the undersurface of the bridge sectional model is the key position for large-scale spanwise vortex formation and shedding. The presence of the ESBM device amplifies the scale of the vortex and increases the aerodynamic force caused by the shedding of the vortex. When installed on the leading edge, the vibration response is reduced, and the control efficiency is 18.0% , which is a positive effect. When installed in the middle, the vertical vibration almost disappears, and the control efficiency reaches 94.1% . It can be considered that the ESBM device installed in the middle position played the role of a central stabiliser, thereby almost completely suppressing the vertical VIV response of the sectional model. When devices are installed symmetrically on the leading and trailing edges, the vibration response is reduced, and the control efficiency reaches 38.6% , which indicates that the effect of the symmetrical installation is better than the effect of placing the device on the leading edge alone. A similar conclusion can also be reached based on the plot of the torsional response. Installation of the ESBM devices on the undersurface of the bridge also acts as a passive flow control measure. It is clear that the leading edge of the undersurface is the key position for suppressing the VIV, and the trailing edge is the worst position.

With the control airflow, compared with the no control state, the F-mode, M-mode, FB-mode, FBX-mode, and FBO-mode completely suppress the vertical VIV of the bridge sectional model, and the control efficiencies are 94.1% , 96.1% , 93.8% , 93.5% , and 93.7% , respectively. When the perturbation mode is the B-mode, there is no effect on the vertical VIV, and the vibrations begin earlier, which is an adverse effect. When the perturbation mode is the FOB-mode, the vertical vibration

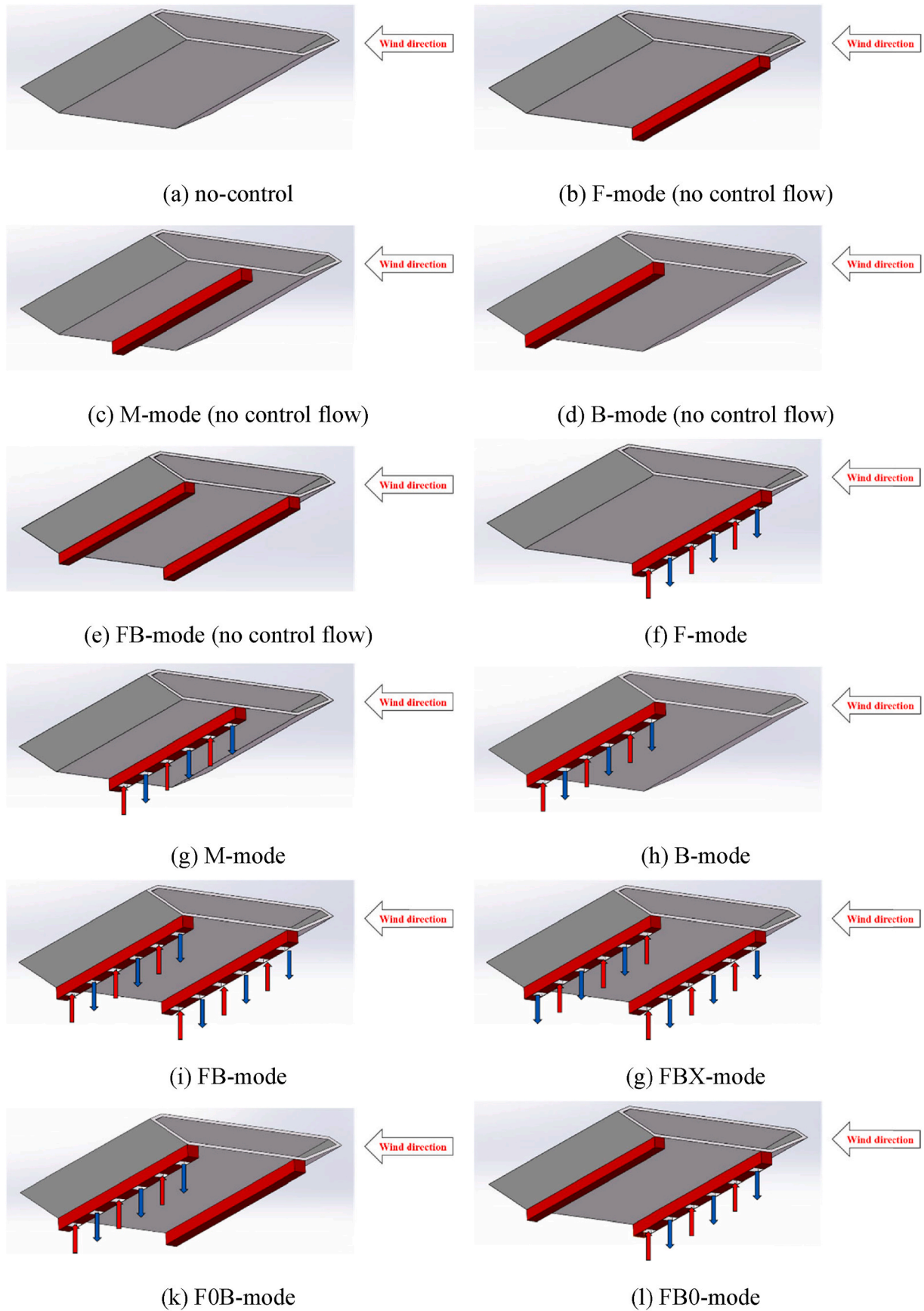


Fig. 7. Perturbation modes.

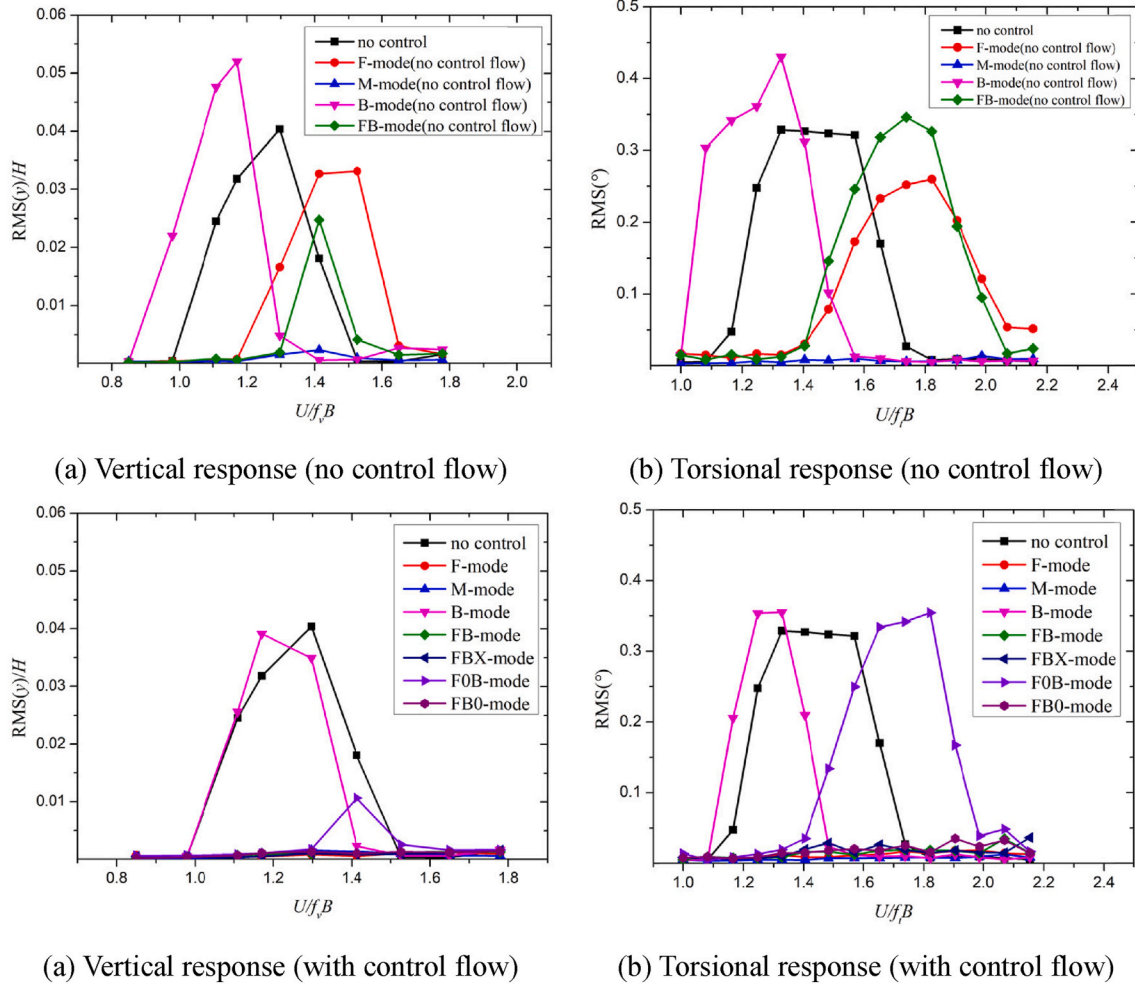


Fig. 8. RMS of the vertical and torsional responses vs. the reduced velocity with different perturbation modes.

decreases, and the control efficiency is 73.6%. Compared with the no control state, the F-mode, M-mode, FB-mode, FBX-mode, and FB0-mode completely suppress the torsional VIV of the bridge sectional model, and the control efficiencies are 94.4%, 96.0%, 93.8%, 93.5%, and 93.6%, respectively. When the perturbation mode is the B-mode, the torsional VIV is not affected, but the vibration is advanced, which is an adverse effect. When the perturbation mode is the F0B-mode, the torsional vibration is not affected, and the vibration is delayed, which is a slight positive effect.

Overall, when the perturbation mode is the B-mode, the device

negatively affects the VIV of the bridge sectional model. The ESBM device installed in the middle of the model undersurface can completely suppress the VIV of the model, regardless of whether it is disturbed. It is demonstrated that when the device plays the role of a [central stabiliser](#), the flow can only increase the control efficiency slightly, and it cannot affect the main role of this passive measure. Similarly, the application of flow cannot offset the adverse effect caused by installation of the ESBM device on the trailing edge of the undersurface. Therefore, when the flow is large enough, it may offset the adverse effects or significantly improve the control effect of the passive control measures. In addition, the

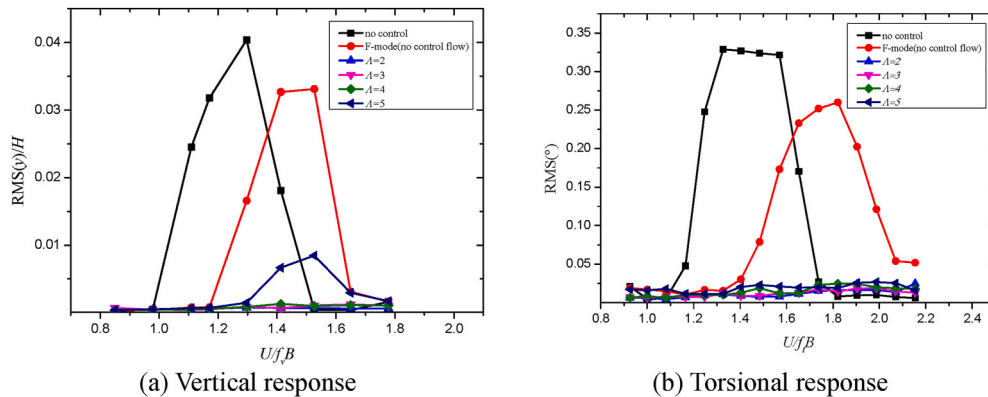


Fig. 9. RMS of the vertical and torsional responses vs. the reduced velocity with different perturbation distances.

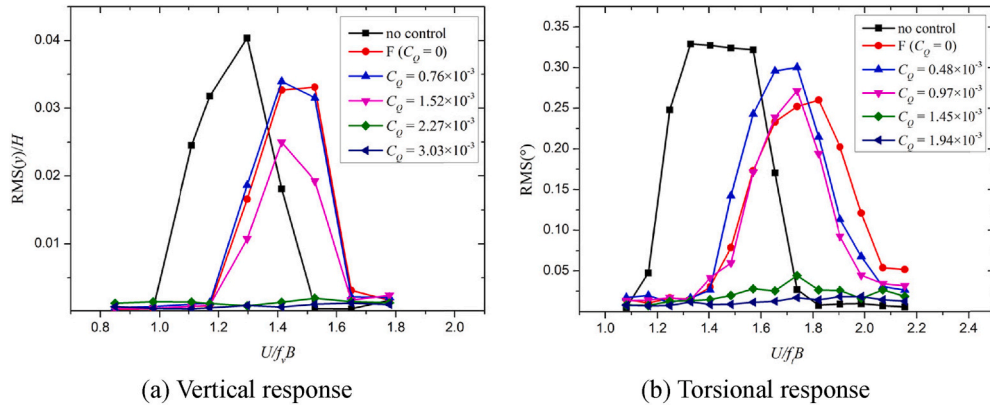


Fig. 10. RMS of the vertical and torsional responses vs. the reduced velocity with different single-hole flow coefficients.

application of a flow does not impact the effect of the passive control measures. When the FB-mode and FBX-mode perturbations are applied, the vibration is both completely suppressed, indicating that whether the air holes on leading and training edges are in the same state does not affect the control effect. The control efficiencies of the FB-mode and FB0-mode are basically the same, which indicates that when the ESBM devices are installed symmetrically, the perturbation of the leading edge is the most important for suppressing the VIV of the bridge sectional model. Moreover, the application of a perturbation at the trailing edge

does not affect the control efficiency, further demonstrating that the leading edge of the undersurface is a favourable perturbation position for the ESBM device, and the trailing edge is unfavourable. Therefore, the F-mode is selected as the perturbation mode for subsequent investigation.

(2) Control effects of the spanwise distance

Under $\alpha = 0$, $C_Q = 3.03 \times 10^{-3}$ for vertical VIV and $C_Q = 1.94 \times 10^{-3}$

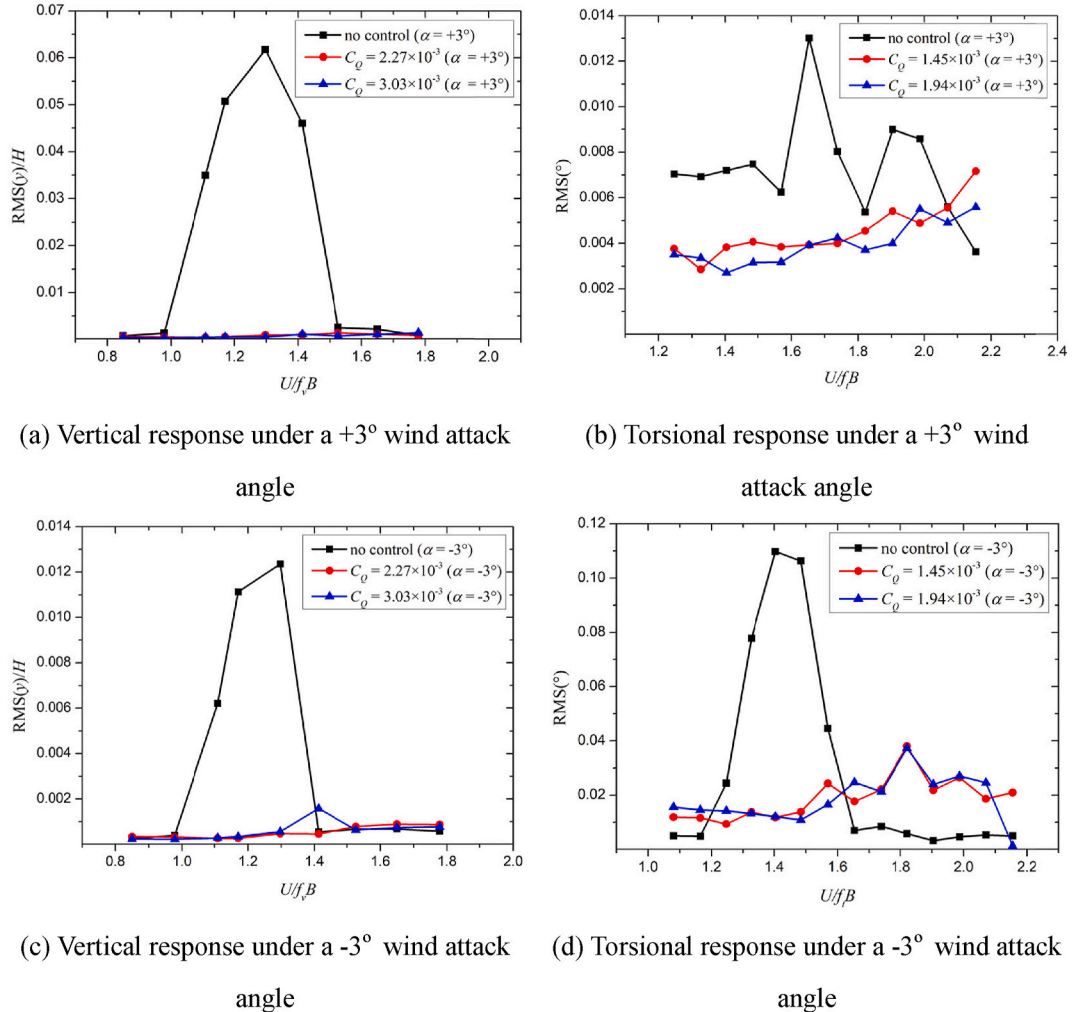


Fig. 11. RMS of the vertical and torsional responses vs. the reduced velocity under different wind attack angles.

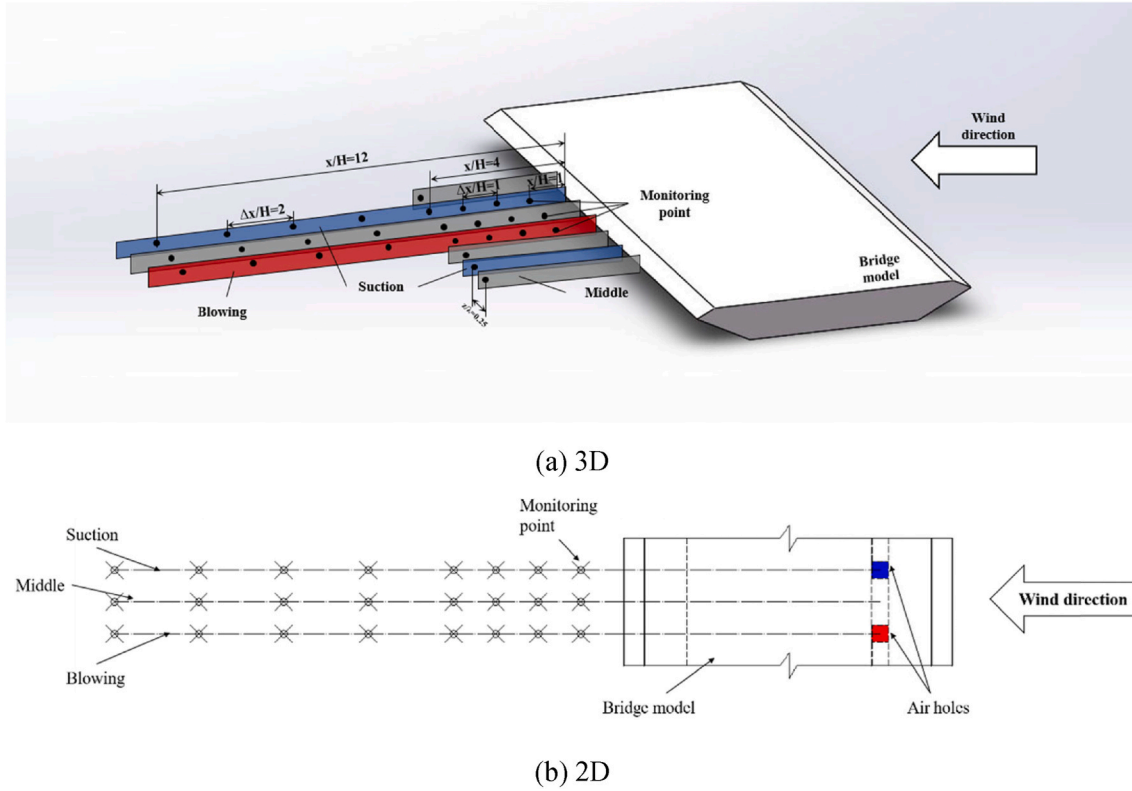


Fig. 12. Distributions of wake monitoring points.

for torsional VIV, the control effects of the devices with the F-mode perturbation for reducing the vortex-induced vertical and torsional responses with different dimensionless spanwise distances are shown in Fig. 9.

It can be seen from the figures that when $\Lambda = 2-4$, the VIV of the bridge sectional model can be effectively suppressed. In the plot of the vertical response, relative to the no control state, the vertical VIV response is effectively suppressed when $\Lambda = 2-4$, and the control efficiencies reach 94.8%, 94.1%, and 92.4%. When $\Lambda = 5$, the RMS of the vertical displacement is reduced, and the control efficiency is 79.0%. In the plot of the torsional response, the torsional VIV response can be completely suppressed when $\Lambda = 2-5$, and the control efficiencies are 92.5%, 94.4%, 92.5%, and 91.8% when the spanwise variation is applied.

(3) Control effects of the single-hole flow coefficient

Under $\alpha = 0$ and $\Lambda = 3$, the control effects of the devices with the F-mode perturbation for reducing the vortex-induced vertical and torsional responses with different single-hole flow coefficients are shown in Fig. 10.

It can be seen from the figures that as the flow increases, the control effect is better. For the vertical response, compared with the no control state, when $C_Q = 0-1.52 \times 10^{-3}$, there is a certain suppression effect on the vertical vibration of the bridge sectional model, and the control efficiencies are 18.0%, 15.9%, and 38.1%. However, the vibration is not fully suppressed. When $C_Q = 2.27 \times 10^{-3}$ and 3.03×10^{-3} , the vibration is basically completely suppressed, and the control efficiencies are 89.4% and 94.1%, respectively. The same conclusions can be reached based on the plot of the torsional response. The control efficiencies with $C_Q = 0-1.94 \times 10^{-3}$ are 20.9%, 8.7%, 17.4%, 79.5%, and 94.4%, and the torsional vibration of the bridge sectional model is basically completely suppressed when $C_Q = 1.45 \times 10^{-3}$ and 1.94×10^{-3} .

(4) Control effects under different wind attack angles

Under $\Lambda = 3$, $C_Q = 2.27 \times 10^{-3}$, 3.03×10^{-3} for vertical VIV and $C_Q = 1.45 \times 10^{-3}$, 1.94×10^{-3} for torsional VIV, the control effects of the devices with F-mode perturbation for reducing the vortex-induced vertical and torsional responses under different wind attack angles are shown in Fig. 11.

It can be seen from the figures that the wind attack angle has a significant influence on the control effect. The ESBM can effectively suppress the VIV of the bridge sectional model under different wind attack angles, even changing the vibration form of the VIV. When $\alpha = +3^\circ$, in the plot of the vertical response, both flow options can completely suppress the vertical vibration of the model, and the control efficiencies are 95.9% and 96.7%. In the plot of the torsional response, compared with the case with a wind angle of attack of zero, the torsional VIV of the bridge sectional model in the no control state basically does not occur, and only random vibration occurs; the control conditions are the same.

When $\alpha = -3^\circ$, in the plot of the vertical response, after the ESBM is applied, both flow options can completely suppress the vertical vibration of the model, and the control efficiencies are 78.1% and 77.9%. In the plot of the torsional response, after the ESBM is applied, both flow options can completely suppress the torsional vibration of the model, the control efficiencies are 55.9% and 82.2%, and only random vibration occurs.

(5) Wake analysis

The wake analysis in this study is divided into spanwise and downstream wake analyses. The 3D and 2D distributions of the wake monitoring points are shown in Fig. 12.

In Fig. 12, the black dots represent the sampling points, which are located in the central section of the suction holes (denoted 'Suction'), the central section of the blowing holes (denoted 'Blowing'), and the section between the two holes (denoted 'Middle'). The sampling points

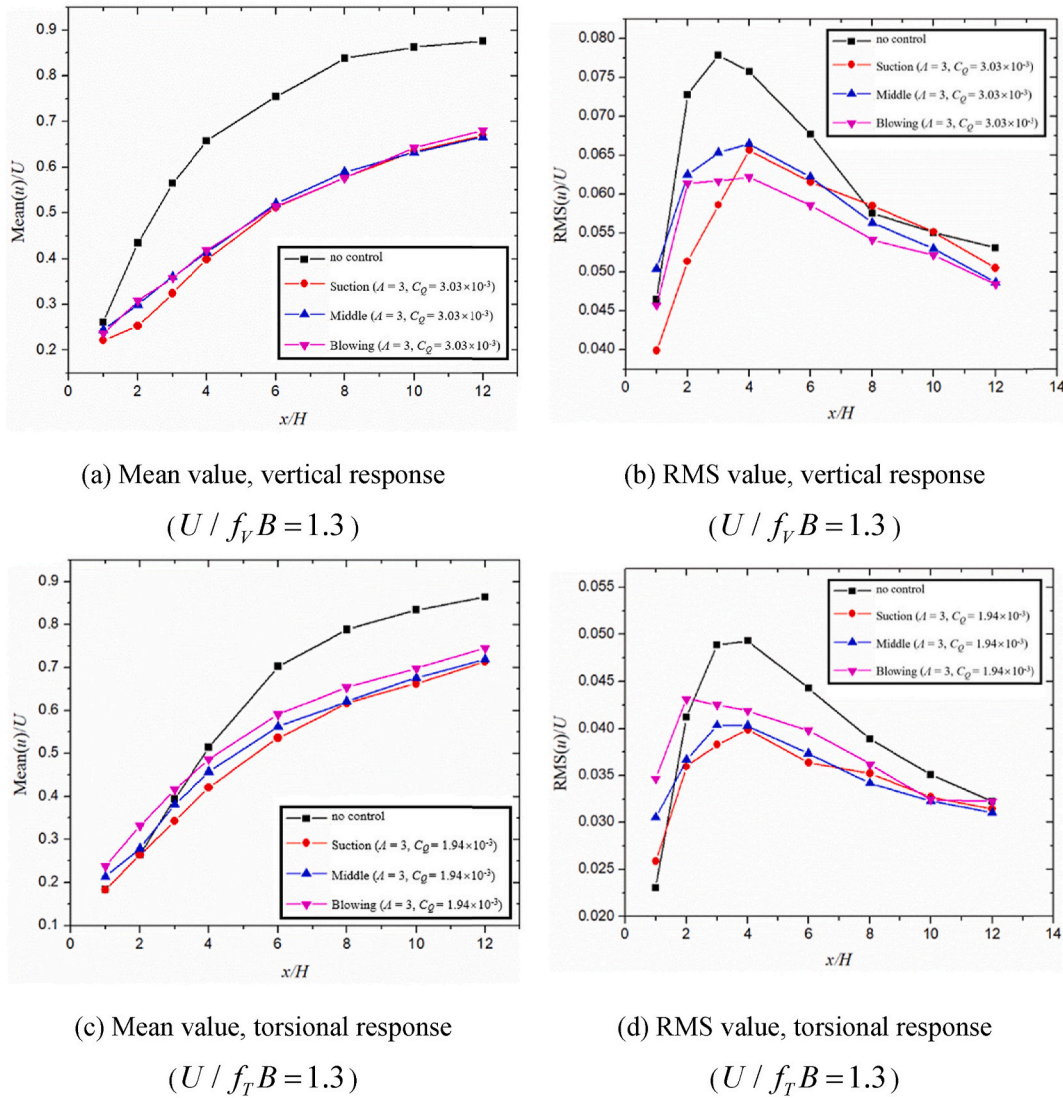


Fig. 13. Evolution of the mean and RMS values of the wind velocity in the downstream direction.

are distributed on the horizontal section located at the trailing edge of the bridge sectional model. There are 24 sampling points distributed in the above three sections to measure the downstream wake evolution, and the spacing is shown in Fig. 12. There are seven sampling points that include a complete suction-blowing cycle to measure the spanwise wake evolution. The control mechanism of the $C_Q = 1.94 \times 10^{-3}$ ESBM is analysed by comparing the wake evolution and the Power Spectral Density (PSD) of the no control and optimal control conditions. The non-dimensional wind speeds $U/f_v B = 1.3$ and $U/f_t B = 1.3$ correspond to the wind speeds with the largest vertical and torsional VIV amplitudes, respectively. Under $\alpha = 0$, $\lambda = 3$, $C_Q = 3.03 \times 10^{-3}$ for the vertical VIV and $C_Q = 1.94 \times 10^{-3}$ for the torsional VIV, with the F-mode perturbation, the evolution of the mean and RMS values of the wind velocity in the downstream direction is shown in Fig. 13.

It can be seen from these figures that the ESBM can reduce the strength of the spanwise vortex, thereby reducing the VIV of the sectional model of the bridge. The mean value of the downstream wind speed increases with the distance from the model. This is because when the measuring point is farther from the model, the influence of the model on the wake is also reduced, which makes the downstream wind speed in the wake closer to the incoming wind speed. Compared with the no control state, the mean values of the three sections corresponding to the control conditions are all smaller than that of the no control state, which

indicates that the application of the ESBM device results in increased aerodynamic drag. As the distance from the model increases, the RMS value initially increases and then decreases. This increase occurs because the scale of the spanwise vortex continues to increase during the shedding process, and the shedding is completed at $x/H = 3$; the subsequent decrease is because the intensity of the spanwise vortex gradually decreases with increasing distance from the model. Compared with the no control state, in the increasing stage, the RMS value of the control condition is smaller than that of the no control state, which confirms that the device can suppress the scale of the vortex. In the decreasing stage, the RMS value of the control condition is smaller than that of the no control state and stabilises earlier, which confirms that the device can accelerate the decay of the spanwise vortex. The differences between the mean and RMS values for the three sections under control conditions are very small, which indicates that the suction/blowing of the ESBM device can effectively suppress the development and shedding of the spanwise vortex structure and can suppress the spanwise vortex farther away from its location.

The evolution of the mean and RMS values of the wind velocity in the spanwise direction are shown in Fig. 14.

Compared with the no control state, the mean and RMS values of the control condition are smaller than those of the no control state, which indicates that the device can suppress the scale of the spanwise vortex.

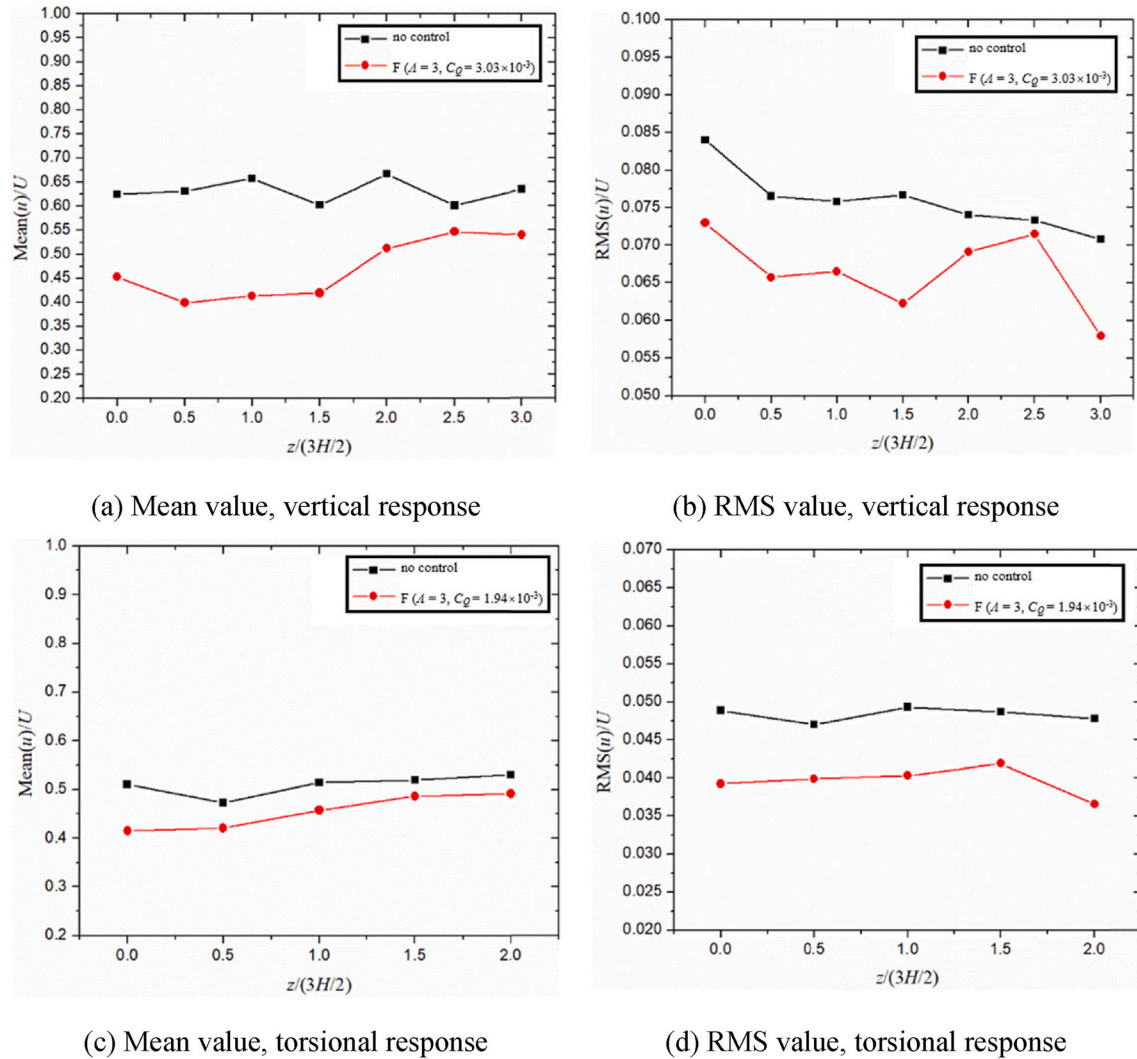


Fig. 14. Evolution of the mean and RMS values of the wind velocity in the spanwise direction.

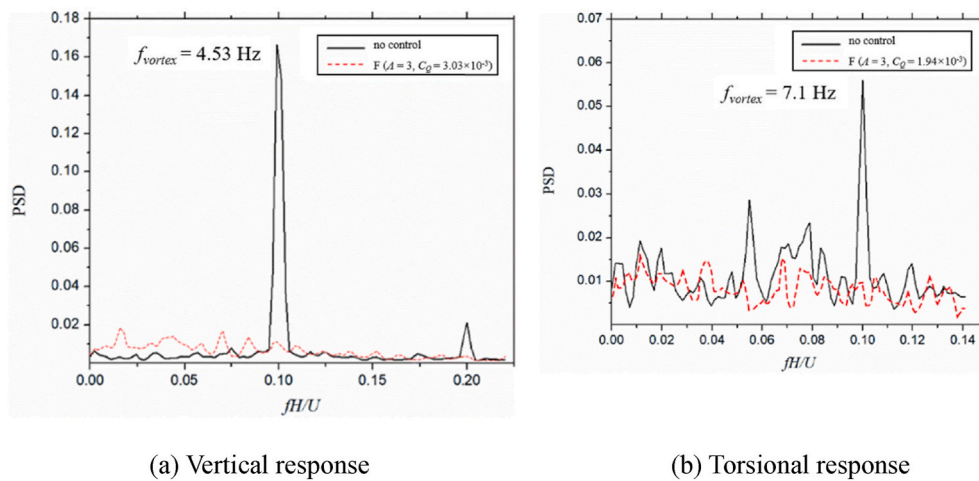


Fig. 15. PSD of the downstream velocity.

The PSD is obtained using the fast Fourier transformation (FFT) of the wake velocity at point $x/H = 4$ located in the central section when vertical and torsional VIV occurs, as shown in Fig. 15.

In both figures, the peak in the PSD at $fH/U = 0.1$ indicates that there

is strong spanwise vortex shedding in the wake, and at the same time, a significant VIV occurs at this wind speed. With the vertical VIV, the frequency of the downstream wind speed is $f_{vortex} = 4.53\text{Hz}$, which is similar to the frequency of the vertical natural vibration of the model,

$f_V = 4.52\text{Hz}$, indicating that the vortex shedding frequency is locked by the vibration frequency at this time. With the torsional VIV, the frequency $f_{\text{vortex}} = 7.1\text{Hz}$ of the downstream wind speed is similar to the frequency of the torsional natural vibration of the model, $f_T = 6.90\text{Hz}$, and the same conclusion is drawn as for the vertical VIV. The PSD peak disappears under the control condition, indicating that the control mechanism of the ESBM suppresses the development and shedding of the spanwise vortex in the wake, thereby suppressing the VIV.

5. Conclusions

The ESBM proposed in this paper is a 3D spanwise-varying control method. The wind tunnel test results demonstrate the influence of different variables of this method on the control effect for VIV. The major conclusions drawn are as follows:

- 1) The ESBM is effective for controlling bridge VIV, particularly when the device is arranged to provide a suction-blowing control flow. The effect is best when the ESBM device is installed on the leading edge of the undersurface of the model. The device installed in the middle plays the role of a central stabiliser. When devices are installed on the leading and trailing edges symmetrically, they can also completely suppress the vibration. The current study uses the F-mode perturbation.
- 2) When the dimensionless spanwise distance is less than 4, and the single-hole flow coefficient is greater than 2.27×10^{-3} during vertical VIV (1.45×10^{-3} during torsional VIV), the suction-blowing flow control from the leading edge causes the VIV to almost disappear.
- 3) The suction-blowing control flow reduces the mean and fluctuating downstream wind speeds in the wake; thus, the scale of the vortex is weakened, which generates the effect of suppressing VIV.

CRedit authorship contribution statement

Zhicheng Xue: Methodology, Writing – original draft, Writing – review & editing. **Bin Han:** Methodology, Validation, Data curation. **Hongfu Zhang:** Methodology, Formal analysis, Investigation. **Dabo Xin:** Conceptualization, Resources, Visualization, Project administration, Funding acquisition. **Jian Zhan:** Writing – review & editing, Investigation. **Rui Wang:** Writing – review & editing, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The support for this work are provided by the National Key Research and Development Program of China (Grant No. 2018YFC0809600, 2018YFC0809605) and the National Natural Science Foundation of China (Grant no. 51878131).

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